Model Comparison and Results

Overview

Our team compared different machine learning models to predict fraudulent insurance claims. We evaluated logistic regression, random forest, and gradient boosting using precision, recall, F1-score, PR-AUC, and the percentage of fraud captured within the top 10% and 20% of highest-risk claims.

We used class weighting to address the imbalance between fraudulent and non-fraudulent claims. All models were evaluated using the same 1995 validation data, with training on 1994 claims.

Model	Precision	Recall	F1	PR-AUC	Top 10%	Top 20%
Baseline logistic regression	0.0556	0.0033	0.0063	0.0981	16.28%	35.22%
Class-weighted logistic regression	0.1009	0.4319	0.1636	0.0948	19.27%	33.89%
Tuned logistic regression	0.1093	0.5282	0.1811	0.1017	17.28%	34.55%
Tuned random forest	0.1144	0.2857	0.1633	0.1099	19.93%	38.87%
Task 3 grid-search random forest	0.1170	0.5050	0.1900	0.1100	20.27%	40.53%
Tuned gradient boosting	0.1098	0.5150	0.1810	0.1417	23.59%	40.53%

Gradient boosting achieved the highest PR-AUC at 0.1417 and captured the highest percentage of fraud within the top 10% of highest-risk claims at 23.59%.

The Task 3 grid-search random forest performed better than the separately tuned random forest in recall and F1-score. It achieved a recall of 0.5050 and an F1-score of 0.1900, compared to 0.2857 and 0.1633 for the tuned random forest. It also captured 40.53% of fraud within the top 20% of highest-risk claims.

Model Selection

Although gradient boosting had the highest PR-AUC and top 10% fraud capture, our Challenge Advisor recommended that random forest was strong enough to move forward with.

Based on this feedback and the results, we plan to continue with the Task 3 grid-search random forest model. It performed competitively across the evaluation metrics and had stronger recall and F1-score than the separately tuned random forest.

Limitations and Next Steps

One limitation we identified is the distribution of fraud across different months. Fraud cases nearly disappear during certain blocks of months that shift between years. In an additional rerun of the models, removing Month and MonthClaimed increased random forest PR-AUC from approximately 0.11 to 0.14. This suggests that the month-related features may be affecting model performance and should be investigated further.

Our next steps are to:

- Continue evaluating the random forest model.
- Investigate the impact of month-related features on fraud predictions.
- Identify which features contribute most to the model's predictions.
- Improve model interpretability and understand the factors associated with fraud risk.
- Prepare the final model documentation and model card.